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Water use and root growth by annual and perennial pastures and subsequent crops in a phase rotation

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Abstract

Perennial pastures and trees have been proposed as a means of increasing water use, reducing groundwater recharge, and therefore reducing the spread of secondary salinity in southern Australia. This paper investigates comparative water use throughout a 5-year rotation involving 3 years of pasture (either annual subterranean clover (*Trifolium subterraneum* L.) or perennial lucerne (*Medicago sativa* L.)) followed by wheat (*Triticum aestivum* L.) and then by canola (*Brassica napus* L.). Within the pasture phase, lucerne reduced soil water content during summer and autumn by approximately 60 mm more than the annual pasture. Lucerne in its second year produced 1.7 times more root biomass than the annual pasture, and also grew roots deeper into the soil. Wheat after lucerne produced 15 t ha^{-1} dry matter compared with 8 t ha^{-1} for wheat after clover, but grain yields were similar due to a combination of frost damage and water stress. There was a small difference in canola dry matter production, but no differences in canola grain yield or 1000-grain weight in the second year after pasture. Canola root distributions were similar on the two blocks. Average annual water leakage was 45 mm for the annual pasture–wheat–canola rotation and 17 mm for the lucerne pasture–wheat–canola rotation. Wheat after lucerne extracted 20 mm more water from the 'B' horizon than wheat after clover, and this extended the period of protection against water leakage. Water leakage under the lucerne block commenced before the dry soil buffer was completely refilled, indicating the possibility of bypass flow. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The major cause of secondary salinity in southern Australia is the removal of native deep-rooted perennial vegetation, and its replacement with shallow-rooted annual crops

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and pastures, resulting in excess groundwater recharge. The causes, extent and costs of secondary salinity in southern Australia have been reviewed in more detail by McFarlane and Williamson (2002). Because the problem operates at the catchment scale, solutions must also be directed at relatively large scales. One possible solution is the adoption of whole-farm plans, integrated within a catchment, incorporating belts of woody perennials, larger areas of herbaceous perennials, and engineering structures to remove excess water. Such a scheme has been implemented on the 'Ucarro' farm (Rundle and Rundle, 2002; Hodgson et al., 2002).

Lucerne (*Medicago sativa* L.) is one of the few perennial pasture plants available for this environment, and it is being actively promoted as one of a range of tools to combat salinity. It is a legume, and fixes nitrogen for use by subsequent crops. Because it is herbaceous, it allows farmers to retain a degree of flexibility in their rotation (e.g. a 3-year lucerne phase followed by a 3-year crop phase), and therefore it is often more acceptable to farmers than large-scale adoption of tree plantations. It is known to extract water from deeper in the soil than annual pastures (Miller et al., 1981; Crawford and Macfarlane, 1995), and uses at least 50 mm more water than annual crops or pastures at many locations across Australia (e.g. Angus et al., 2001; Dunin et al., 2001; Latta et al., 2001; McCallum et al., 2001; Ridley et al., 2001; Ward et al., 2001). Ward (1986; cited by Dexter, 1991) found that lucerne seedlings were able to penetrate compact soil layers, whereas wheat seedling roots were deflected. However, there are no direct observations of root distribution for lucerne under Australian conditions. There are only inferences based on the depth of water uptake.

The additional water use conferred by lucerne is comparable in magnitude with average excess groundwater recharge, estimated for the wheat belt of southwestern Australia as 20–50 mm per year (Nulsen, 1998). Lucerne is therefore capable of significantly influencing the quantity of groundwater recharge in areas where it is well adapted to local edaphic and climatic conditions.

However, the effects of a phase of lucerne on subsequent water leakage and crop performance are less well understood. Does the dry soil left by lucerne completely refill before water leakage commences, or does bypass flow result in leakage before the buffer is filled? Does a dry 'B' horizon adversely affect crop performance in some years, as suggested by the modelling of McCallum et al. (2001)? Alternatively, can subsequent crops use the root channels presumably created by lucerne in otherwise inhospitable soils to gain access to greater stored soil water? This could result in increased crop yields, as the cropping regions of southern Australia are frequently subjected to terminal droughts. If greater soil exploration and water use by crops after lucerne does occur, does this provide an extended period of protection from groundwater leakage beyond the root zone?

In this paper we aim, firstly, to compare root distributions for annual and lucerne pastures on a duplex soil in southwestern Australia. We then examine the impact of pasture type on subsequent crop performance, in terms of both crop production and the timing and quantity of water leakage from the root zone. Finally, we compare total water leakage from the 3 years annual pasture/crop/crop rotation with water leakage from the 3 years lucerne pasture/crop/crop rotation.

2. Methods

2.1. Site

The site was located 10 km west of Katanning, in southwestern Australia (Hodgson et al., 2002). The soil was a yellow sandy duplex soil (Typic Palexerult—USDA Soil Taxonomy), with approximately 40 cm loamy sand overlying a red massively structured medium clay. At a depth of approximately 80 cm, the structure changed to blocky angular, and at about 120 cm, a zone of white clay began. The soil surface sloped at about 3°. The experimental area was bounded at the top and bottom by drains cut to a depth of 0.7 m, with three rows of trees planted immediately downslope from the drain. Trees and drains were installed approximately 10 years previously on a 1:300 grade. The average annual rainfall at the town site is 483 mm, with 362 mm falling in the May–October period. Further details have been given by Hodgson et al. (2002).

2.2. Pastures

Subterranean clover (*Trifolium subterraneum* L. cv. Dalkeith and Seaton Park; 4 ha) and lucerne (cv. Sceptre; 7 ha) pastures were sown on adjacent blocks in June 1995. The clover pasture was set stocked according to district practice, and the lucerne pasture was rotationally grazed approximately 1 week in 7, commencing in December 1995. In 1996 and 1997, both pastures were sprayed to control grass weeds and both became legume-dominant. The lucerne density was approximately 30 plants m^{-2}. Dry matter production from the two pastures was approximately equal for each 12-month period (Lisa Blacklow, personal communication, 1999). In March and April 1998, rainfall was 116 mm (decile 9), and the lucerne grew to approximately 40 cm high, before both pastures were sprayed with a mix of glyphosate (0.9 kg a.i. ha^{-1}) and 2,4-D-ester (0.8 kg a.i. ha^{-1}) on 26 April 1998. The lucerne block (but not the clover block) was scarified to a depth of 10 cm 2 weeks later. These treatments resulted in a 95% kill of the lucerne.

2.3. Wheat

Wheat (*Triticum aestivum* L. cv. Cunderdin) was sown at 60 kg ha^{-1} on 11 May 1998, with 7.6 kg P ha^{-1} and 17.5 kg N ha^{-1}. A further 18.4 kg N ha^{-1} as urea was applied on 16 August 1998, and sub-plots receiving different amounts of N were also established at this time (see below). Dry matter production was assessed in ten randomly located 1.0 m^2 quadrats in each main block on 23 July, 23 September (coinciding with the first signs of anthesis) and at maturity on 4 December. Grain yield, 1000-grain weight, and straw and grain N were also measured on the samples taken at maturity.

Response to N fertiliser was assessed in triplicate plots (10 m $\times$ 10 m) arranged randomly in a transect on each block from the top of the block to the bottom, approximately 20 m from the boundary between the two main blocks. Each plot received 0, 18.4, 36.8 or 55.2 kg N ha^{-1} as urea. Dry matter production, grain yield, 1000-grain weight and grain nitrogen were assessed on samples taken on 4 December from a 1.0 m^2 quadrat located randomly in each plot.

2.4. Canola

Canola (*Brassica napus* L. cv. Pinnacle) was sown on 13 May 1999 at a rate of 5.8 kg ha^{-1} with 7.6 kg P ha^{-1} and 54.3 kg N ha^{-1}. Dry matter production was assessed in ten randomly located 1.0 m^2 quadrats in each main block on 22 September and at maturity on 8 December. Grain yield and 1000-grain weight were also recorded for samples taken at maturity.

2.5. Soil water content

Soil water content was monitored in each block by both time-domain reflectometry (TDR) at one location (near hole no. 22 in the clover plot and no. 23 in the lucerne plot, see Hodgson et al., 2002) at eight depths (0–10 cm vertically with 10 cm long probes, and then horizontally at 15, 25 and 35 cm soil depth with 40 cm long probes, and then at 50, 70, 90 and 110 cm soil depth with 20 cm long probes), and neutron moisture meter (NMM) at 10 locations to 175 cm. The NMM access tubes used were numbers 9–18 in the clover plot, and numbers 27–36 in the lucerne plot (Hodgson et al., 2002). TDR measurements were recorded at 15 min intervals, and NMM readings were obtained at approximately monthly intervals. TDR measurements commenced in 1995, before sowing lucerne and clover pastures. Throughout the 3-year pasture phase, NMM measurements, integrating water content changes across the blocks, were compared with simultaneous TDR measurements. TDR readings were then adjusted so that they closely matched NMM responses. Further details are given by Ward et al. (2001). NMM readings alone were used to monitor soil water content changes for soil depths between 1.1 and 1.9 m. Available water in the ‘A’ horizon was calculated assuming that the water storage measured at the time of senescence of the annual pasture corresponded with available water of 0 mm. In the ‘B’ horizon, minimum measured soil water storage under the lucerne was assumed to correspond to available water of 0 mm. The ‘C’ horizon was assumed to have the same water holding capacity, per unit of soil depth, as the ‘B’ horizon.

2.6. Evapotranspiration

Bowen ratio energy balance equipment (McIlroy, 1972) was installed on each block to monitor total evapotranspiration (transpiration plus soil evaporation) throughout the growing season at 30 min intervals. Where possible, gaps in the data from one of the units were filled from a correlation, calculated separately for each month, for data from the other unit. For the few periods when both units were inoperative, gaps were filled from a correlation with net radiation and vapour pressure deficit (Ward and Dunin, 2001).

2.7. Water leakage

Water leakage was calculated from the water balance equation $D = P - \mathrm{ET} - \Delta S$, where D is the water leakage, P the rainfall, ET the evapotranspiration, and ΔS is the change in soil water storage. Although lateral water flow collected by the drains was monitored, it was always less than 5 mm per year, and was not included in our water balance calculations.

2.8. Pasture and crop root distribution

Root growth distributions were assessed for both pastures on 23 October 1996, using 15 cm diameter cores obtained using a split-barrel auger system to a depth of 2.1 m. Ten cores were taken from each main block, in a transect from the upper tree rows to the lower ditch. The cores were separated at depths of 0.1, 0.2, 0.3, 0.45, 0.6, 0.8, 1.0, 1.2, 1.5, 1.8 and 2.1 m. Each soil sample was washed over a 2.0 mm sieve, and roots were collected, dried at 60 °C, and weighed.

In addition, three soil pits were dug with a backhoe to a depth of approximately 1.5 m on 10 September 1997 in each block. Maximum rooting depth was assessed qualitatively in each pit by spraying the sides of the pit with water and carefully excavating individual roots.

In the canola crop, root distribution was measured on soil samples from 10 randomly selected locations in each main plot on 22 September 1999. At each location, samples of 10 cm diameter were taken in 10 cm increments with a hand auger to a depth of 50 cm, and a petrol-powered auger for depths from 50 to 100 cm.

3. Results

3.1. Soil water content

Soil water contents under the pastures have been described by Ward et al. (2001). Briefly, soils under lucerne were dried by an additional 60 mm compared with the annual pasture (Fig. 1). The difference occurred during summer and autumn, with water drawn mainly from soil below a depth of about 60 cm. The difference was reduced, sometimes to zero, during winter, but was re-established each summer. Water uptake by lucerne extended to a depth of approximately 1.7 m (data not shown).

In May 1998, when wheat was sown, soil previously under lucerne was considerably drier than soil previously under clover (Figs. 1 and 2). However, by late August, soil water contents to a depth of 1.2 m in the two blocks were similar. Water contents in the two blocks then followed similar patterns as they dried, until about the middle of October. At this point, both TDR and NMM indicated that soil water contents under wheat after lucerne declined further than for wheat after clover. This effect was most marked at soil depths greater than 0.7 m, and was maintained throughout the summer. A large rainfall event in March 1999 reduced the differences between the plots.

During the 1999 growing season, when canola was grown, soil water contents once again quickly converged, and thereafter followed the same patterns in each block until monitoring finished in December 1999 (TDR) or April 2000 (NMM). At the end of monitoring, there was little difference in soil water content between the two plots, except that the lucerne block was still about 10 mm drier between soil depths of 1.2 and 1.9 m (Fig. 2).

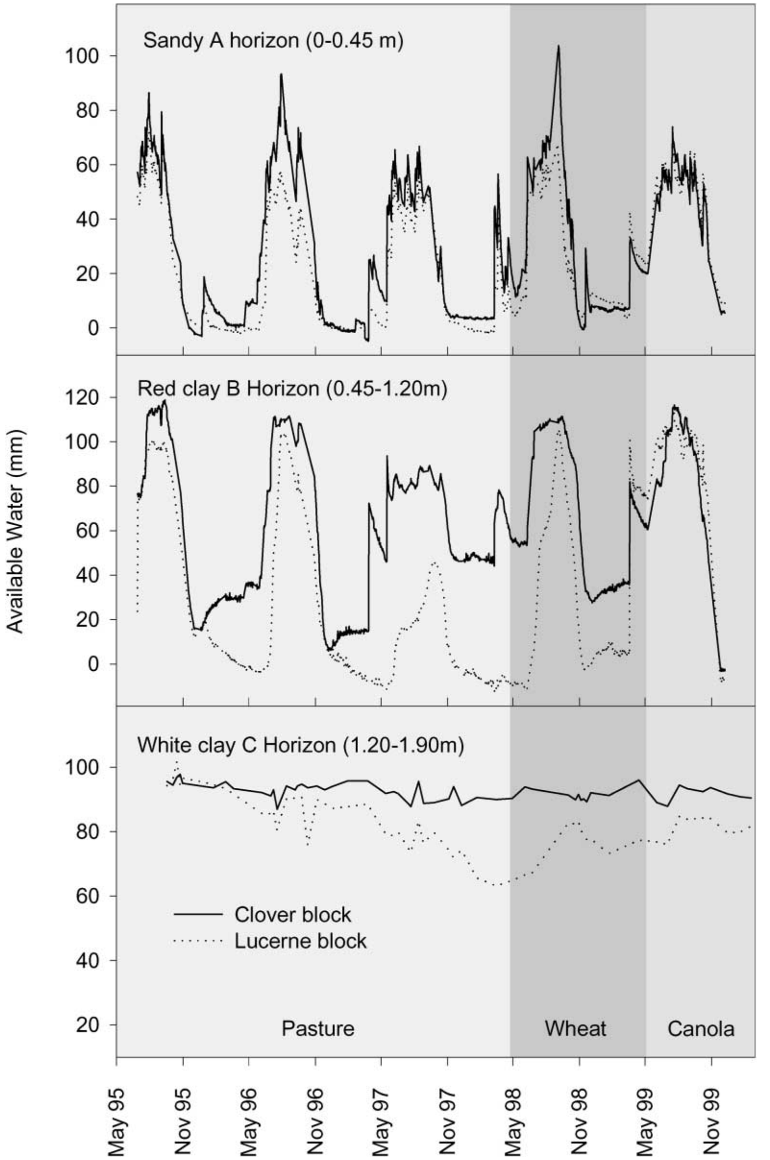

Fig. 1. Available water storage in the 'A', 'B' (TDR) and 'C' (NMM) horizons in the clover and lucerne blocks throughout the 5-year rotation.

3.2. Evapotranspiration

Actual evapotranspiration by the wheat and canola crops followed the same patterns as for lucerne and clover at this site (Ward et al., 2001). Total ET closely matched potential ET (Priestley and Taylor, 1972) from the break of the season (April or May) until soil

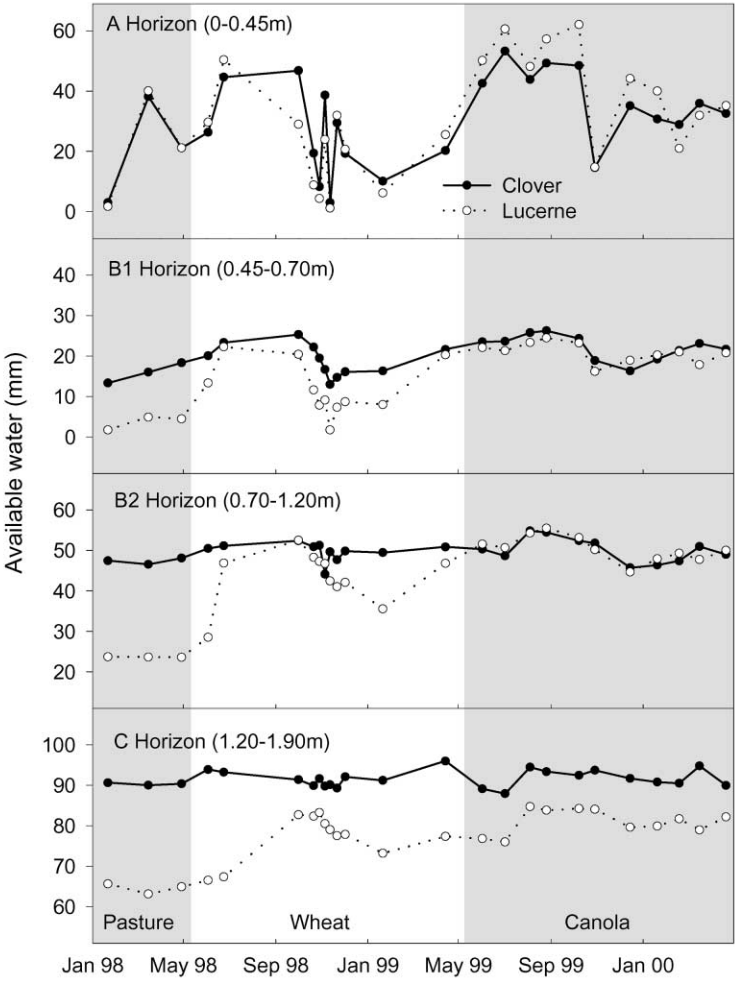

Fig. 2. Available water storage (NMM) in the 'A', 'B1', 'B2' and 'C' horizons under wheat (1998) and canola (1999) in blocks previously under clover or lucerne.

water became limiting in September or October, depending on the seasonal conditions (Fig. 3). After this date (approximately 6 October 1998 for wheat on the lucerne block; 12 October 1998 for wheat on the clover block; 24 September 1999 for canola on the lucerne block; and 3 October 1999 for canola on the clover block), actual evapotranspiration diverged from potential evapotranspiration. For the wheat crop (1998), no differences in evapotranspiration could be detected by Bowen ratio energy balance between the two plots, despite the apparent differences in soil water use detected by both NMM and TDR. However, as discussed by Ward et al. (2001) and Angus and Watts (1984), the Bowen ratio energy-balance technique can become prone to large errors under conditions of low vapour flux and high available energy, such as occur during late spring and summer (commencing in November) in this environment.

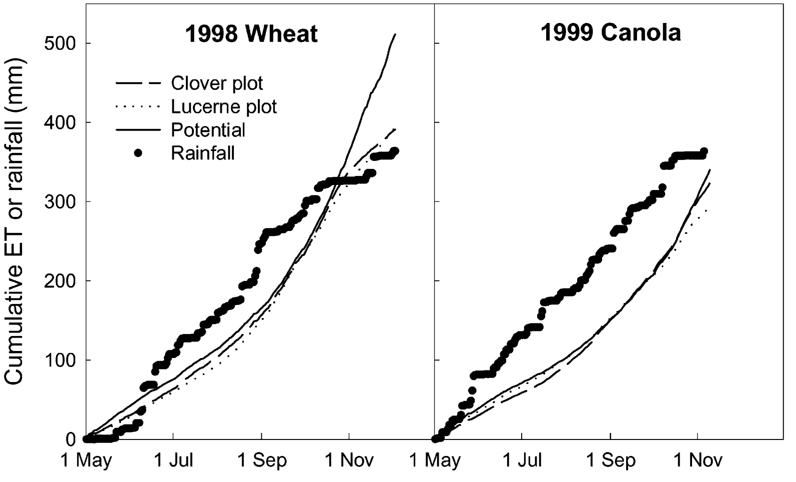

Fig. 3. Cumulative rainfall and actual and potential evapotranspiration during the May–November period from wheat (1998) and canola (1999) in blocks previously under clover or lucerne.

3.3. Pasture root distribution

The second-year lucerne stand (sampled in 1996) contained more root material at all depths compared with the annual pasture (Fig. 4), despite having similar levels of above-ground production. Integrated over the whole profile, soil under the lucerne pasture

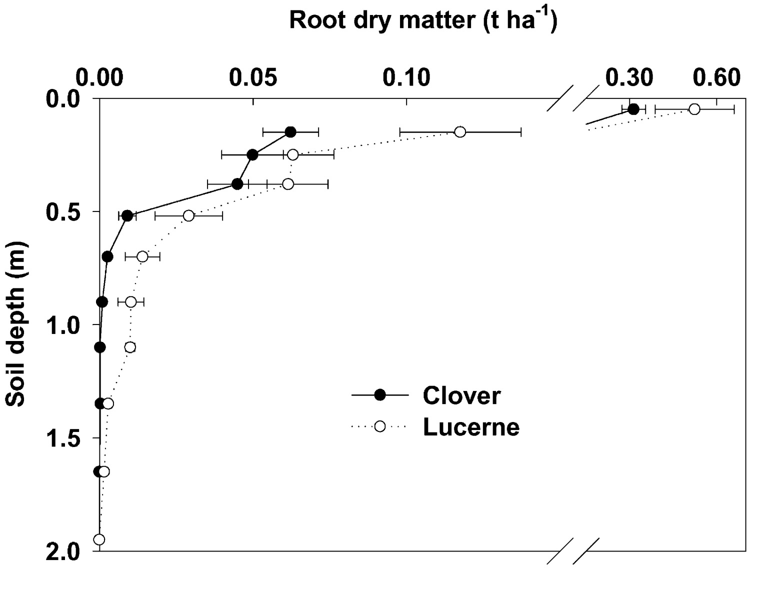

Fig. 4. Comparative root distributions for subterranean clover (annual) and lucerne (perennial) pastures in their second year. Error bars represent 1 S.E. ($n = 10$).

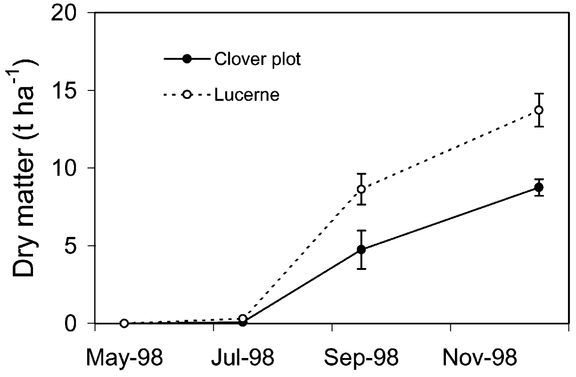

Fig. 5. Dry matter accumulation by wheat after either clover or lucerne. Error bars represent 1 S.E. ($n = 10$).

contained 1.7 times the root mass under the annual pasture. The majority of roots of both pastures were contained in the top 10 cm, and root density decreased rapidly with depth, particularly below a depth of 50 cm, which corresponds with the boundary between the sandy 'A' horizon and the clay 'B' horizon. Very few clover roots were found at soil depths greater that 0.7 m, but lucerne roots were recovered from below 1.5 m.

Soil pits dug in 1997 (third-year lucerne) showed considerably greater rooting depth for the lucerne compared with the clover pasture. Lucerne roots were traced to the bottom of the soil pits (approximately 1.5 m), whereas roots under the annual pasture could not be detected below approximately 0.5 m, which agrees with the root distributions measured in the previous year. However, the actual root distribution could not be quantified from these observations.

3.4. Crop growth, root distribution and yield

Wheat (1998) after lucerne performed considerably better at 14 t ha^{-1} than wheat after clover (8 t ha^{-1}), in terms of dry matter production (Fig. 5). However, mild frost (Bronte Rundle, personal communication, 2000) and a dry spring prevented the yield potentials from being realised, so that grain yields were less than 4 t ha^{-1} for both blocks (Fig. 6). In the plots receiving different rates of nitrogen fertiliser, wheat after clover responded more to nitrogen (in terms of dry matter production) than wheat after lucerne (Fig. 6a). Trends in grain yield were similar to those for dry matter production (Fig. 6a), but of smaller magnitude due to the late frost mentioned above. Grain nitrogen content (Fig. 6b) was higher for wheat after lucerne, but grain size was smaller (Fig. 6c).

In the canola crop, there was a significant difference in dry matter production, but differences in grain yield and 1000-grain weight were non-significant (Table 1). Similarly, there were no major differences in root distribution (Fig. 7), although for soil depths between 0.45 and 0.65 m, canola on the clover plot produced more roots than canola on the lucerne plot. Root distribution on both plots was dominated by tap roots in the 0–10 cm soil depth. In comparison with the pasture root distribution (Fig. 4), canola had much higher total root biomass in the 0–10 cm range, although as different sampling techniques

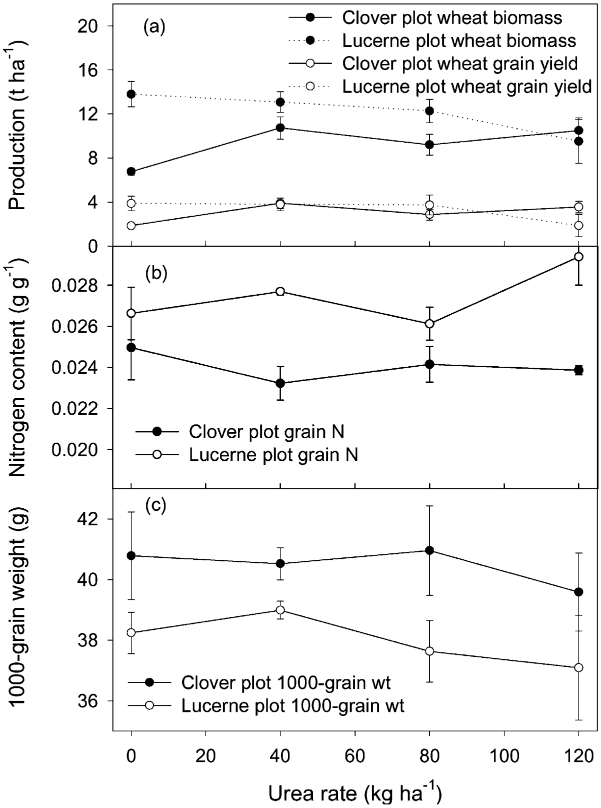

Fig. 6. Influence of previous pasture type and urea application rate on wheat (a) dry matter production and grain yield, (b) grain nitrogen content, and (c) 1000-grain weight. Error bars represent 1 S.E. ($n = 3$).

were used, the results are not strictly comparable. Both canola and pasture roots showed evidence of accumulation at the base of the sandy 'A' horizon (about 30–50 cm).

3.5. Water leakage

In 1995, water leakage was similar for the clover and the establishing lucerne plot (Table 2). However, in subsequent years, even after removal of the lucerne in April 1998,

Table 1
Dry matter production and grain yield for canola grown 2 years after lucerne or clover pasture[a]

	Dry matter (t ha^{-1})	Grain yield (t ha^{-1})	1000-grain weight (g)
Clover plot	13.3	2.5	2.80
Lucerne plot	15.0	2.9	2.91
Probability	0.034	0.425	0.221

[a] Probability values were calculated from an unpaired t-test assuming equal variances.

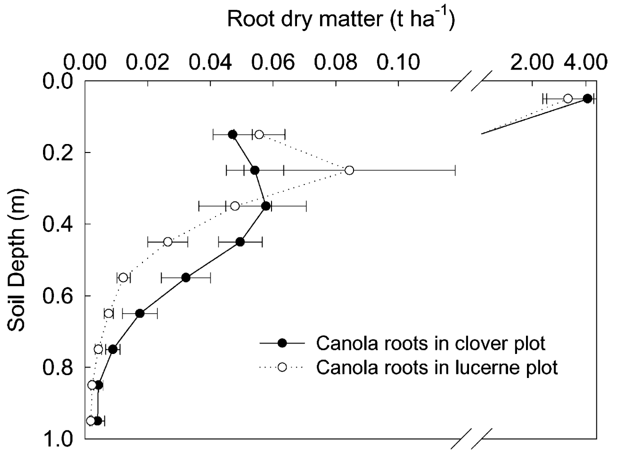

Fig. 7. Comparative root distributions for canola grown 2 years after either subterranean clover (annual) or lucerne (perennial) pasture. Error bars represent 1 S.E. ($n = 10$).

Table 2
Water leakage below 1.9 m for the clover and lucerne plots throughout the rotation, for the growing season (May–October), and for a full 12-month period (January–December)

Year	Clover		Lucerne	
	May–October	January–December	May–October	January–December
1995	10	10	25	25
1996	65	65	10	10
1997	25	25	−20	−20
1998	15	25	−10	−10
1999	80	100	80	80
Average	39	45	17	17

water leakage was significantly less in the lucerne plot, reflecting the greater soil water deficit created by the deeper roots of lucerne. Due to unseasonal rainfall, some water leakage was detected in the autumn (February–April) of 1999 and 2000, and these figures were also included in Table 2 to present a true picture of the impact of the dry soil buffer created by lucerne.

4. Discussion

4.1. Crop production and root growth

In 1998, the wheat crop following lucerne produced far more biomass than the wheat following the annual pasture, despite both pastures being managed for maximum legume component (about 80% by visual estimate) and minimum grass component (less than 10%, again by visual estimate). Results from the sub-plots receiving different rates of nitrogen fertiliser suggest that the major reason for the difference was the level of available nitrogen. Protein content in the grain was also significantly greater for wheat after lucerne than for wheat after clover. Previous studies (Holford and Crocker, 1997; Holford et al., 1998; Peoples et al., 1998; McCallum et al., 2000) have concluded that lucerne pastures fix more nitrogen than comparable annual legume pastures, mainly because lucerne pastures usually have a greater legume component than traditional annual pastures. In some instances, the extra nitrogen fixed by lucerne reduced the nitrogen fertiliser requirement for subsequent crops for several years (Holford and Crocker, 1997; Holford et al., 1998). However, in our experiment, there was little difference in above-ground legume production between the two pastures. Differences could have been generated by the differences measured in root distribution, particularly when combined with the cultivation before sowing for the wheat after lucerne but not the wheat after clover, but we have no data to confirm this.

One fear amongst the farming community of introducing lucerne into a crop rotation is that the dry soil left by lucerne may impact adversely on crop yields, particularly in an environment characterised by a terminal drought. Climatic conditions in 1998 were such that the difference in stored soil water at the break of the season in May was large (about 65 mm), and the last 2 weeks of October and the first 2 weeks of November were dry (about decile 2 for the 4-week period). These conditions increase the risk of a more severe terminal drought induced by lucerne. However, because late August 1998 was particularly wet (71 mm between 17 and 30 August), the soil in the lucerne block wet up to the same level as the soil in the clover plot, so both wheat crops had potential access to the same quantity of water. The greater biomass for wheat after lucerne resulted in faster water use during October and November as determined from soil water contents (Figs. 1 and 2), but this could not be confirmed by Bowen ratio measurements, which become unreliable at this time of year (Angus and Watts, 1984). In our experiment, there was no difference in crop yield, despite the considerable differences in biomass during the dry phase in October and November. There was, however, an impact on grain size, with grain size for wheat after lucerne being significantly smaller, suggesting that the crop was more stressed than wheat after clover, possibly as a result of the considerably greater biomass. Under drier seasonal conditions, where the 'B' horizon was not replenished, the greater biomass of the wheat after lucerne could result in significant water stress during grain filling.

Soil water content data from the neutron probe indicated that wheat after lucerne extracted water from as deep as 1.7 m, compared with 0.7 m for wheat after clover. Wheat presumably used root channels created by lucerne, which enabled the wheat after lucerne to extract an extra 20 mm of water from deep soil layers not available to wheat after clover. The fact that this did not follow through to an increase in grain yield was probably due to frost. However, the extra water use may have had an impact on water leakage in 1999 (see next section).

For the canola crop (second year after pasture), dry matter production on the lucerne plot was greater than on the clover plot, and canola after clover produced more roots in the top of the 'B' horizon. However, these differences were small in magnitude, and there were no differences between the two main plots in terms of crop yield or 1000-grain weight. In contrast to the results of Holford and Crocker (1997), there appeared to be no increase in nitrogen availability 2 years after the lucerne pasture, relative to the clover pasture, although any inherent differences between blocks may have been obscured by the nitrogen applied as fertiliser (54 kg N ha^{-1}).

The similar root distributions for the two blocks could be because any deep root channels created by lucerne, and subsequently used by wheat, were unavailable for the canola crop. The root channels themselves may have closed, or for some reason the presence of the old wheat roots may have prevented canola root growth. The latter option is unlikely, as there are no reports in the literature of wheat having an inhibitory effect on canola root growth. Therefore, the most likely scenario is that the lucerne root channels had closed, possibly through natural soil swelling or translocation of clay particles.

4.2. Water leakage in the rotation

There is no doubt now that lucerne uses more water than conventional annual crops and pastures in many situations in southern Australia (Carbon et al., 1982; Crawford and Macfarlane, 1995; Dunin et al., 1999; Ward et al., 2001). The root distributions reported here confirm that lucerne produced more roots than a comparable annual pasture, and also penetrated deeper into the soil, allowing the observed greater water use from deeper soil layers. The resulting greater depth of dry soil acted as a storage zone for subsequent rainfall, reducing the amount of water leaking beyond the root zone and into the groundwater. The effect of lucerne became apparent in its first summer, and continued throughout the lucerne phase (Ward et al., 2001). Although differences between the two main plots in topography, and possibly in soil type, may have influenced the quantities of water leakage (Hatton et al., 2002), there is no doubt that pasture type was also a major factor.

At the conclusion of the lucerne phase, we expected that during the cropping phase, the total difference in water leakage between the two plots would be 60 mm (the difference in soil water storage at the start of the cropping phase). This is close to what actually happened, with total leakage during 1998 and 1999 on the block previously in lucerne of 80 mm, and 130 mm on the block previously in clover. However, in 1999, leakage occurred from the lucerne block before the dry soil storage zone was completely filled. Even after the 1999 growing season, and more unseasonal rain in December 1999 and March 2000, there was still a difference in stored soil water of 10 mm for soils between 1.2 and 1.9 m (Fig. 2). This suggests that bypass flow may be occurring through the ‘B’ horizon. In other words, water leakage happens through the larger pores, bypassing some of the smaller pores in the soil matrix. We anticipate that the residual 10 mm dry soil buffer will gradually be expressed as reduced water leakage during the next few years.

Because of the possible bypass flow mechanism, extra water uptake by wheat in 1998 did not influence water leakage during the 1999 growing season. However, for the full 12 months of 1999, water leakage from the lucerne plot was less than for the clover plot, reflecting both the effect of lucerne and also the effect of extra water uptake by wheat. However, the lack of response by canola roots in the second year after the pastures suggests that the influence of deep root channels on subsequent crops is likely to be short-lived. More research is necessary to clarify the influence of lucerne root channels on root growth of subsequent crops, and the impact of this on the soil water balance.

5. Conclusions

Lucerne, in contrast to clover, used considerably more water from deep soil layers, and over a 5-year rotation (3 years lucerne, wheat, canola) reduced average annual water leakage from 45 to 17 mm. This equates to a 60% decrease in potential groundwater recharge, and according to George et al. (1999), should considerably delay the onset of secondary salinity, and perhaps significantly reduce the eventual extent. Long-term modelling is required to determine the impact of a phase of perennial pasture on the average rate of water leakage. Water leakage, calculated from the water balance, began on the lucerne block before the soil water contents increased to pre-lucerne levels, suggesting the possibility of bypass flow as a significant groundwater recharge mechanism. Enhanced water uptake was observed for the first crop following lucerne (wheat), which may act to extend the life of the dry soil buffer created by lucerne. There were no differences in water uptake under the second crop (canola), probably because the root channels created by lucerne had closed.

Wheat grown in the first year after lucerne (on cultivated soil) produced far more dry matter than wheat after clover (on uncultivated soil). This was probably due to greater nitrogen availability, either through greater N stored by lucerne roots, or an effect of the soil cultivation prior to sowing, or a combination of both. However, due to a combination of frost damage and a dry period during late October and early November, there was no difference in grain yield. In the second year after pasture, there was a small difference in canola dry matter production, but no differences in canola, grain yield, 1000-grain weight or root growth.

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